

Robert J. Spencer

AI Engineer · PhD Researcher in AI for Epidemiology

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AI Engineer and PhD Researcher delivering production ML systems and peer-reviewed research concurrently. At Radix Nutrition I built AI/ML optimisation systems that cut formulation iteration time by 90%+ and formulation costs by 25%+. First-author Q1 publication in *Energy and Buildings* (2025). Now researching privacy-preserving digital twins for antimicrobial resistance (AMR) risk assessment at The University of Queensland and the SAAFE CRC.

SKILLS

Languages	Python, SQL, R, JavaScript/TypeScript
Machine learning	PyTorch, scikit-learn, transformer models, time-series forecasting, transfer learning, optimisation
Data & deployment	Data pipelines, Pandas, Polars, Docker, CI/CD, PyQt, HPC
Research focus	Digital twins, synthetic data, federated learning, One Health / AMR risk assessment

EXPERIENCE

Beyond Data
AI Engineer (Contract)

Brisbane, Australia
Jun 2026 – Present

Delivering end-to-end data and AI solutions for Australian businesses, working across the full delivery lifecycle alongside doctoral research.

- Translating client requirements into solution architecture, then building and deploying the data pipelines and ML systems that support it.
- Working across data engineering, ML development, and deployment as a single thread of ownership rather than a hand-off between specialists.

The University of Queensland & SAAFE CRC
PhD Researcher

Brisbane, Australia
Apr 2026 – Present

Researching cross-sectoral digital twins for antimicrobial resistance (AMR) risk assessment in agribusiness, food, and environmental systems.

- Developing privacy-preserving digital twin methods that integrate biological, environmental, agribusiness, and food-system data under a One Health framework.
- Applying synthetic data generation, federated learning, and transfer learning to address data scarcity and fragmentation across distributed data environments.
- Running scalable machine learning and simulation workloads on UQ's Bunya high-performance computing (HPC) infrastructure.

Radix Nutrition
Senior Data Scientist

Hamilton, New Zealand
Jan 2025 – Oct 2025

Led the end-to-end design, development, and deployment of AI-driven solutions, acting as both product owner and individual contributor.

- Built proprietary AI/ML optimisation systems (Python, PyTorch) integrating multiple nutritional databases (NZFCD, AFCD, USDA FoodData Central) and encoding complex FSANZ Food Standards Code logic, cutting formulation iteration time by 90%+ (from weeks to days). Presented live demonstrations of this work to [Google & Fonterra](#).
- Led data-driven substantiation for all general level health claims, applying the FSANZ framework to quantify 150+ unique health benefits for a single product line and enabling key marketing positions such as 'World's Healthiest'.
- Co-developed AI-optimised breakfast and smoothie formulations, reducing formulation costs by 25%+ while modelling for compliance across nutrient targets and multiple front-of-pack (FoP) labelling systems, including Nutri-Score, HSR, NPSC, and traffic-light labels.
- Developed in-house tools – dashboards, GUIs, scripts, and ad hoc analyses – adopted by R&D and Operations to accelerate cross-team decision-making.

Radix's first technical hire, responsible for establishing the company's data and software foundations.

- Bootstrapped company-wide data infrastructure, analytics workflows, and internal web tools supporting R&D and operations.
- Worked hands-on across data analytics, software engineering, web development, and applied nutrition in a high-velocity environment.

Massey University

Albany, Auckland, New Zealand

Research Assistant, Artificial Intelligence

Sep 2023 – Oct 2024

Transformers, transfer learning, and time-series forecasting for building energy prediction.

- As first author, led the project end-to-end: co-defined the research direction, conducted experimentation across 16 Building Data Genome 2 datasets, and authored the manuscript, published in *Energy and Buildings* (Q1, 2025).
- Achieved a 15.9% reduction in MAE for 24-hour forecasting using a multi-source dataset approach.
- Implemented and compared multiple Transformer architectures, demonstrating PatchTST outperformance versus vanilla Transformer and Informer, and developed both zero-shot and fine-tuning transfer-learning pipelines.

EDUCATION**The University of Queensland** · Doctor of Philosophy (Artificial Intelligence)

2026 – 2029 (expected)

- Fully funded RTP Scholarship and SAAFE CRC top-up.
- Supervisors: Dr Noorul Amin and Professor Ricardo Soares Magalhães.

Monash University · Master of Computer Science (Artificial Intelligence)

2024 – 2027 (expected)

- GPA 3.83/4.0 (High Distinction average).
- Part-time and remote, undertaken alongside full-time work and research.
- Completed 9 of 12 units; remaining units planned concurrently following doctoral confirmation.

Massey University · Master of Analytics (Health)

2023 – 2024

- GPA 8.17/9.0, awarded with Distinction.
- Research stipend funded by a Massey University MURF Seed grant, supporting ML approaches for occupant comfort and energy performance in commercial buildings.

The University of Waikato · Bachelor of Health, Sport and Human Performance

2020 – 2022

- Double Major in Human Performance Science and Community Health.

PUBLICATIONS

Spencer, R., Ranathunga, S., Boulic, M., van Heerden, H., & Susnjak, T. (2025). Transfer learning on transformers for building energy consumption forecasting—A comparative study. *Energy and Buildings* (Q1). doi.org/10.1016/j.enbuild.2025.115632